HARDWARE LIST FOR OPEN SOURCE LAB

The examples and prices shown were discovered through Internet searches. The total cost is $15,912. It is likely costs can be reduced substantially:

- The school district may already have compatible hardware that can be used in the lab.
- The district may be able to negotiate lower prices through its vendor relationships and through buying in bulk.
- The district may be able to solicit donations from companies or individuals.

The proposed thin-client computers and LCD monitors would result in a substantial energy savings. Each thin client, including the LCD monitor, uses fewer than 33 watts. A typical desktop machine, including a CRT monitor, uses at least 200 watts. Thus, a thin client results in an 83.5% energy savings over a typical desktop machine.

Proportional to the amount of energy used is the amount of heat generated. The thin clients would generate substantially less heat, which would not only make the room more comfortable, but also save money on air conditioning.

Finally, because the thin clients make no noise, the classroom environment will be more conducive to learning.

Description	Example	Features/Notes	Cost	Quantity	Subtotal
Server computer—these are the machines that are actually doing the work; the students log into them from their thin clients. To do 3D graphics, these machines must be powerful.	HP ProLiant BL465c Server (http://tinyurl.com/2hsbs4): 	• 2 Opteron processors to enable 26 clients to do 3D animation • 6 GB memory to assure that processes run fast	$2,614	2	$5,228
Thin-client computers—these are the computers students will be using in the lab. Because they are thin clients, they have no disk drive; students cannot install software on them or reconfigure them. Indeed, they require no configuration whatsoever; just plug them into the network and turn them on, and they will run off of the Edubuntu server.	Norhtech "Microclient Sr" (http://www.norhtec.com/products/mcsr): 	• Uses less than 10 watts of power (less than 20% of a typical PC) • Silent (there is no disk drive or fan, accordingly, no moving parts) • Mountable on the back of an LCD monitor • Generates very little heat	$195	26	$5070

Description	Example	Features/Notes	Cost	Quantity	Subtotal
LCD monitors	ViewSonic VA503B (http://tinyurl.com/5f7ur6): 	• Uses 23 watts of power (less than 20% of a typical PC) • Generates very little heat	$186.50	26	$4849
PS/2 keyboards	iMicro KB-819EB PS/2 107-Key Keyboard (http://tinyurl.com/5who57): 	• Small footprint leaves more room on table for students to work, as well as makes it easier to put a group of students together at a table	$4.99	26	$129.74
PS/2 optical mice	Logitech Value Optical Mouse SBF90 (http://tinyurl.com/6p8kvx): 	• Three-button mouse with scroll wheel is required for 3D animation software • Optical mouse makes maintenance much easier as there are no track balls to lose or get dirty	$5.44	26	$141.44
Headphones—students will use these when recording and editing audio files	Sony MDR-V150 Monitor Series Headphones (http://tinyurl.com/6ygxgj): 	• Over-the-ear design makes it easier for a student to edit audio in a noisy classroom	$17.49	26	$454.74
Microphone—students will use these for recording sound effects and dialog	Labtec Verse 504 Desktop Microphone (http://tinyurl.com/5w6by3): 	• Any inexpensive microphone with a 1/8" jack will do	$1.49	26	$38.74

CURRICULUM MAP FOR OPEN SOURCE LAB AT MIDDLE SCHOOL

Applicable Standards

Performance Standard for Applied Learning	Description
A1 Problem Solving	Apply problem-solving strategies in purposeful ways, both in situations where the problem and desirable solutions are clearly evident and in situations requiring a creative approach to achieve an outcome.
A2 Communication Tools and Techniques	Communicate information and ideas in ways that are appropriate to the purpose and audience through spoken, written, and graphic means of expression.
A3 Information Tools and Techniques	Use information-gathering techniques, analyze and evaluate information, and use information technology to assist in collecting, analyzing, organizing, and presenting information.
A4 Learning and Self-Management Tools and Techniques	Manage and direct one's own learning.
A5 Tools and Techniques for Working with Others	Work with others to achieve a shared goal, help other people to learn on-the-job, and respond effectively to the needs of a client.

National Center on Education and the Economy, **New Standards Performance Standards, Vol. 2, Middle School,** pp. 112-16 (1997).

Desired Result

Enduring Understandings	Essential Questions	Knowledge	Skills	Resources
Layout, color, and design determine the tone and usability of a web page.	• How do different layout choices make a web page easier or harder to understand? • What are the immediate impressions conveyed by particular layout, color, and design choices? • What are the different kinds of fonts and what kind of message do they convey?	• Basic principles of web design. • Cartesian coordinates • HTML/CSS • Color theory • Basic principles of typography	• Web-page layout. • Using Cartesian coordinates to describe a layout. • Hand coding web pages and style sheets • Choosing appropriate colors for backgrounds and text. • Editing fonts	Graph paper Pencils Crayons Text Editor Screem Agave TuxPaint FontForge
Background images provide patterns that affect the message conveyed by web pages.	• What photographic subjects make good backgrounds? • How is color represented on a computer? • What is the dynamic range of an image? • What are appropriate pixel and file sizes for an image for a web page? • How are images included in HTML and CSS code?	• How lighting and framing affect a photograph. • HTML and CSS code for including images.	• Photography • Levels adjustments and RGB histograms • Cropping and resizing • Including images in the background of a web page	Digital cameras G.I.M.P. Screem

Enduring Understandings	Essential Questions	Knowledge	Skills	Resources
A web server is a computer that stores web pages and transmits them across a network on request.	• How are web pages served? • Where are web pages stored?	• Terms and concepts relating to networks, the web, and computer storage.	• Properly naming and storing files so that they may be viewed as web pages over the in-class network.	Nautilus
Animation creates the illusion of fluid motion through the display of a series of images.	• How does animation work?	• Terms and concepts relating to animation, including key frames, tweening, and story boards	• Creating a story board. • Making a flipbook animation. • Making a simple 2D animation on the computer.	Paper Pencils PostIt Notes G.I.M.P.
Three-dimension (3D) animation is the creation of 2D images based upon a 3D model.	• How are objects described and situated in three dimensions? • How are complex objects created by combining simpler ones? • How do lights and colors interact in a 3D scene?		• Creation and manipulation of simple 3D models. • Combination and grouping of objects. • Setting colors of objects. • Placing and focusing lights.	Blender

Enduring Understandings	Essential Questions	Knowledge	Skills	Resources
Audio is represented on a computer as a series of numbers, which can be manipulated to change the nature of the sound.	• What are sound waves? • How are sound waves digitized for a computer? • How are sound waves manipulated on a computer? • How do sound waves relate to frequencies?	• Terms and concepts relating to sound waves. • Sampling	• Using a microphone to record sounds. • Cutting, looping, and adjusting audio volume and other parameters. • Analyzing an audio file to determine its primary frequencies and other characteristics. • Saving an audio file in a format that can be heard on the web.	Microphone Headphones Audacity
Computer hardware transmits, stores, and converts electronic signals to and from input and output.	• How do computers work?	• Terms and concepts relating to the parts of computers and how they work.		Old desktop computer system that can be disassembled
Choosing appropriate names and folder locations for files makes it easier to find them later.	• How are files and folders organized on a computer?	• File organization.	• Traversing and exploring folder hierarchies using a file manager. • Renaming, copying, creating aliases, and moving files. • Using a command-line interface to navigate a file system and manipulate files.	Nautilus Terminal

Enduring Understandings	Essential Questions	Knowledge	Skills	Resources
An operating system controls the interaction between devices and processes on a computer.	• What kind of information does an operating system need to keep track of? • How are processes organized and controlled on a computer?		• Viewing the state information of a Linux system through the etc and proc folders. • Viewing information about various processes using terminal commands.	Terminal
A computer program is a set of unambiguous instructions for how to do something, including how to respond to events.	• What is an algorithm?	• Terms and concepts relating to algorithms, decisions, looping, input, and output.	• Write a computer program. • Troubleshoot and improve it.	Guido van Robot KTurtle
A computer program can simulate real-world physics by responding to the event of a clock ticking.			• Design a Blender game where objects respond to gravity, each other, and user input.	Blender
Spreadsheets are useful tools for organizing numbers and making computations and graphs.			• Entering data into cells. • Moving, copying, filling, and formatting cells. • Creating formulas. • Creating graphs.	OpenOffice

Acceptable Evidence

Performance Tasks	Work Samples	Observations and Dialogues

Performance Tasks	Work Samples	Observations and Dialogues
Web Page Layout and Design • Design layout for web page. • Choose color scheme. • Choose fonts. • Create and include background images. • Create a "template" web page showing all of the style elements.	Published web page. Notes to other team members on how they might improve their web pages. Oral report by team leaders.	Teacher observes ongoing student work. One member of each five-student team is responsible for collecting student work from the team. Each class ends with a wrap-up session where participation is sought from all.
Web Page on Celestial Objects • Research constellations, planets, and other celestial objects that can be seen from different parts of the world, using the Celestia program available on Edubuntu. • Create a web page describing your findings. • Make the web page user friendly and aesthetically pleasing.	Published web page. Team members' notes. Leaders' oral reports.	Teacher observes ongoing student work. One member of each five-student team is responsible for collecting student work from the team. Each class ends with a wrap-up session where participation is sought from all.
Web Portfolio/2D Animation • Create a cover page for a web portfolio of work done in this class. • Link the cover page to the astronomy page. • Create a new page showing how a 2D animation can be created using G.I.M.P. and the resulting animation.	Web portfolio with three web pages. Team members' notes. Leaders' oral reports.	Teacher observes ongoing student work. One member of each five-student team is responsible for collecting student work from the team. Each class ends with a wrap-up session where participation is sought from all.
3D Animation • Create a 3D animation using Blender. • Put it on a web page and include it in the web portfolio.	Blender file including scene. Movie file including animation. Web page included in portfolio showing animation process and resulting animation. Team members' notes. Leaders' oral reports.	Teacher observes ongoing student work. One member of each five-student team is responsible for collecting student work from the team. Each class ends with a wrap-up session where participation is sought from all.
Audio Production • Create an audio play. • Write a script. • Make sound effects. • Include music. • Record and edit the dialog using Audacity. • Include the finished version in the web portfolio.	Written script for audio play. Audio files containing sound effects and music. Finished audio play as an audio file included in a web page.	Teacher observes ongoing student work. One member of each five-student team is responsible for collecting student work from the team. Each class ends with a wrap-up session where participation is sought from all.

Performance Tasks	Work Samples	Observations and Dialogues
Report on Computer Systems • Create a new web page describing how computer systems work, including images and animations where appropriate.	Web page showing how computer systems work.	Teacher observes ongoing student work. One member of each five-student team is responsible for collecting student work from the team. Each class ends with a wrap-up session where participation is sought from all.
Programming I • Write a computer program to escape a maze using Dr. Guido Van Robot. • Write a computer program to draw interesting shapes using Logo. • Include a web page in the portfolio describing how the programs work, the challenges faced, and opinions regarding the pros and cons of the different programming environments.	Computer programs. Web page showing screen shots of the programs, code excerpts, and a description of how the programs work.	Teacher observes ongoing student work. One member of each five-student team is responsible for collecting student work from the team. Each class ends with a wrap-up session where participation is sought from all.
Programming II • Design a Blender game where objects respond to gravity, each other, and user input. • Create a web page showing screen shots of the game and describing it.	Blender file containing scene and physics. Web page showing screen shots and describing project.	Teacher observes ongoing student work. One member of each five-student team is responsible for collecting student work from the team. Each class ends with a wrap-up session where participation is sought from all.
Spreadsheets • Analyze given data, create graphs to visualize the analysis, and include the analysis on a web page.	OpenOffice spreadsheet containing computations, analysis, and graphs. Web page showing screen shots from spreadsheet and describing project.	Teacher observes ongoing student work. One member of each five-student team is responsible for collecting student work from the team. Each class ends with a wrap-up session where participation is sought from all.